

SESSION 11 • LECTURE 9

Unsteady Flow Processes — Filling and Emptying of Vessels

Detailed lecture notes — transient balances, charging/discharging derivations, four solved numericals

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Learning objectives. After this session you should be able to: (i) write the unsteady (transient) mass and energy balances; (ii) derive $u_2 = h_{line}$ for charging an insulated evacuated vessel and $T_2 = \gamma T_{line}$ for an ideal gas; (iii) analyse discharge of a vessel; (iv) solve filling/emptying numericals.

1. When Steady Flow Fails

Filling a gas cylinder, discharging an air receiver, starting a boiler — in all these the mass and stored energy of the control volume change with time. The analysis integrates the balances over the interval between initial state 1 and final state 2 of the CV.

$$(m_2 - m_1)_{CV} = \Sigma m_{in} - \Sigma m_{out}$$

Unsteady mass balance

$$Q - W = (m_2 u_2 - m_1 u_1)_{CV} - \Sigma m_{in} h_{in} + \Sigma m_{out} h_{out}$$

Unsteady energy balance (stream KE, PE neglected)

Uniform-flow idealisation: the state of the fluid at each inlet/exit is constant in time (e.g. h_{in} fixed by a large supply line), so the flow integrals reduce to simple products $m \cdot h$.

2. Charging an Insulated, Evacuated Vessel

1. Rigid, insulated tank, initially evacuated ($m_1 = 0$), connected through a valve to a large line at p_{line} , T_{line} .
2. Open the valve; gas flows in until the tank pressure equals the line pressure; no gas leaves ($m_{out} = 0$).
3. $Q = 0$ (insulated); $W = 0$ (rigid boundary); mass balance gives $m_{in} = m_2$.
4. Energy balance: $0 = m_2 u_2 - m_2 h_{line} \Rightarrow u_2 = h_{line}$.
5. Ideal gas: $c_v T_2 = c_p T_{line} \Rightarrow T_2 = \gamma T_{line}$.

KEY POINT The gas in the tank ends up HOTTER than the supply line (for air, $T_2 = 1.4 T_{line}$): the flow work $p v$ done in pushing each kilogram in is converted to internal energy. Note that T_2 is independent of the final pressure.

3. Discharging a Vessel

- Now $m_{in} = 0$, $m_{out} > 0$; energy balance: $Q = (m_2 u_2 - m_1 u_1) + m_{out} h_{out}$, with h_{out} at the (falling) tank state.
- For rapid, adiabatic discharge the gas remaining inside behaves as though it expanded isentropically: $T_2/T_1 = (p_2/p_1)^{(\gamma-1)/\gamma}$ — the tank **cools**.
- Everyday evidence: a deodorant can or tyre valve turns cold on rapid discharge.
- Remember the pair as opposites: charging heats, discharging cools.

4. Worked Examples

WORKED EXAMPLE 9.1 — Filling an evacuated tank

An insulated, rigid, evacuated tank is connected to an air line at 600 kPa, 300 K and filled until the pressure equalises. Take $\gamma = 1.4$. Find the final temperature.

Solution.

1. Charging result: $u_2 = h_{\text{line}} \Rightarrow T_2 = \gamma T_{\text{line}}$.
2. $T_2 = 1.4 \times 300 = 420 \text{ K} = 147^\circ\text{C}$.

ANSWER $T_2 = 420 \text{ K} (147^\circ\text{C})$

WORKED EXAMPLE 9.2 — Mass admitted during filling

An insulated, rigid, evacuated bottle of 0.05 m^3 is filled from a line at 500 kPa, 290 K ($R = 0.287 \text{ kJ/kg}\cdot\text{K}$, $\gamma = 1.4$). Find T_2 and the mass of air that enters.

Solution.

1. $T_2 = \gamma T_{\text{line}} = 1.4 \times 290 = 406 \text{ K}$.
2. Final state: $p_2 = 500 \text{ kPa}$, $V = 0.05 \text{ m}^3$, $T_2 = 406 \text{ K}$.
3. $m_2 = p_2 V / (RT_2) = (500 \times 0.05) / (0.287 \times 406) = 25/116.5 = 0.2146 \text{ kg}$.

ANSWER $T_2 = 406 \text{ K} \mid m \approx 0.215 \text{ kg}$

Note: $pV = mRT$ with p in kPa, V in m^3 , R in $\text{kJ/kg}\cdot\text{K}$ gives m in kg directly.

WORKED EXAMPLE 9.3 — Charging a tank that already contains gas

A rigid, insulated 0.2 m^3 tank contains air at 100 kPa, 300 K. It is charged from a line at 800 kPa, 320 K until the tank reaches 800 kPa ($\gamma = 1.4$, $c_v = 0.718$, $c_p = 1.005$, $R = 0.287$). Find the final temperature and the mass admitted.

Solution.

1. Initial mass: $m_1 = p_1 V / (RT_1) = (100 \times 0.2) / (0.287 \times 300) = 0.2323 \text{ kg}$.
2. Energy balance ($Q = W = 0$, $m_{\text{out}} = 0$): $m_2 u_2 - m_1 u_1 = (m_2 - m_1) h_{\text{line}}$.
3. Ideal gas (measure u , h from 0 K): $m_2 c_v T_2 = m_1 c_v T_1 + (m_2 - m_1) c_p T_{\text{line}}$, with $m_2 = p_2 V / (RT_2)$.
4. $m_2 T_2 = p_2 V / R = (800 \times 0.2) / 0.287 = 557.5 \text{ kg}\cdot\text{K}$ — so the LHS is fixed: $0.718 \times 557.5 = 400.3 \text{ kJ}$.
5. RHS: $0.718 \times 0.2323 \times 300 + (m_2 - 0.2323) \times 1.005 \times 320 = 50.04 + 321.6(m_2 - 0.2323)$.
6. Solve: $400.3 = 50.04 + 321.6 m_2 - 74.71 \Rightarrow 321.6 m_2 = 424.97 \Rightarrow m_2 = 1.3214 \text{ kg}$.
7. $T_2 = 557.5 / m_2 = 557.5 / 1.3214 = 421.9 \text{ K}$; mass admitted $= m_2 - m_1 = 1.089 \text{ kg}$.

ANSWER $T_2 \approx 422 \text{ K} \mid m_{\text{in}} \approx 1.09 \text{ kg}$

Note: Because $m_2 T_2 = p_2 V / R$ is known, the balance becomes linear in m_2 — a standard exam device.

WORKED EXAMPLE 9.4 — Rapid discharge cooling

An air receiver at 800 kPa, 400 K discharges rapidly until the pressure falls to 400 kPa. Treating the remaining air as expanding reversibly and adiabatically ($\gamma = 1.4$), estimate its temperature.

Solution.

1. $T_2/T_1 = (p_2/p_1)^{(\gamma-1)/\gamma} = 0.5^{0.2857}$.
2. $0.5^{0.2857} = e^{-0.6931 \times 0.2857} = e^{-0.198} = 0.8203$.
3. $T_2 = 400 \times 0.8203 = 328 \text{ K}$ — a drop of 72 K.

ANSWER $T_2 \approx 328 \text{ K} (55^\circ\text{C})$

5. Practice Problems (answers given)

#	Problem	Answer
P1	An insulated evacuated vessel is filled from a nitrogen line at 350 K ($\gamma = 1.4$). Find the final gas temperature.	490 K
P2	Repeat Example 9.2 for a 0.08 m^3 bottle and a line at 600 kPa, 300 K.	$T_2 = 420 \text{ K}$; $m \approx 0.398 \text{ kg}$

P3	A tank at 1 MPa, 350 K discharges to 250 kPa rapidly ($\gamma = 1.4$). Estimate the final temperature inside.	$\approx 235.5 \text{ K}$
P4	Why does the filled tank not depend on how fast the valve is opened? (two lines)	End states fixed by the balance, not the rate

6. Summary

- Unsteady balances: $(m_2 - m_1) = m_{in} - m_{out}$; $Q - W = (m_2 u_2 - m_1 u_1) - m_{in} h_{in} + m_{out} h_{out}$.
- Insulated, rigid, evacuated charging: $u_2 = h_{line}$; ideal gas $T_2 = \gamma T_{line}$ (hotter than the line, independent of p).
- Partially filled tanks: use $m_2 T_2 = p_2 V/R$ to make the balance linear.
- Rapid discharge cools the remaining gas: $T_2/T_1 = (p_2/p_1)^{(\gamma-1)/\gamma}$.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 5 (First Law Applied to Flow Processes — variable flow section). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.